Retrieving Data with SELECT

Lecture Notes · Game DB Junior

In this lesson

- How to get data out of a table with **SELECT** and **FROM**
- What the star * means
- How to choose only **some columns**, change their order, or repeat them

1. Getting all the data

We now have some information in the cities table. Next, we need to **fetch** (retrieve) it. A very simple way to pull information out of a table is:

```
SELECT * FROM cities;
```

When we run it, we see the four cities we added so far.

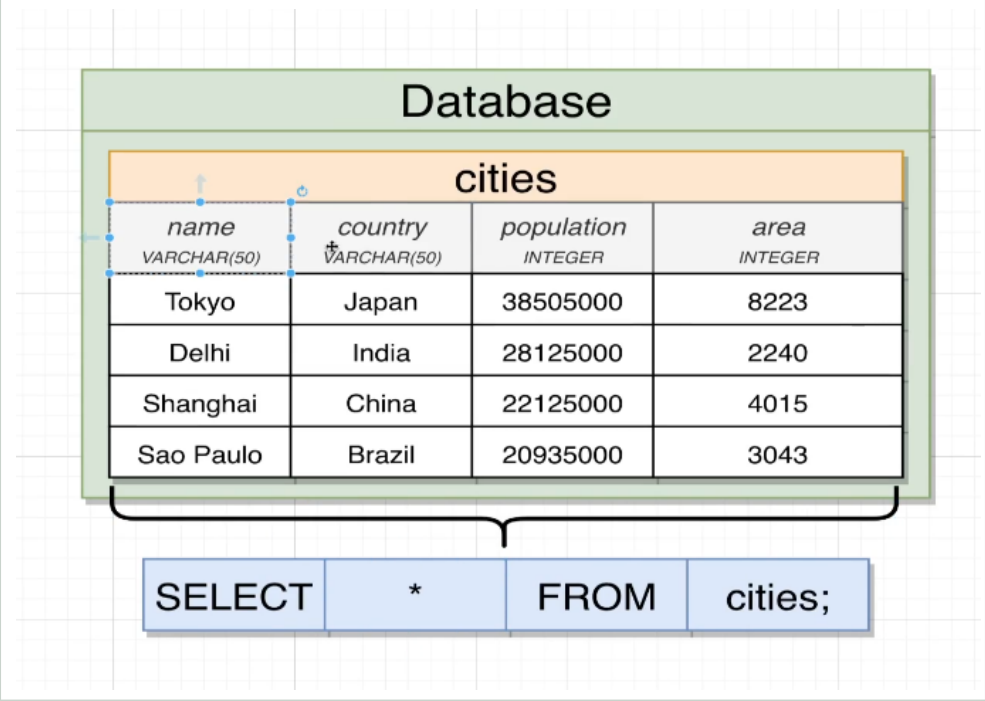


*Figure 1 — SELECT * FROM cities; returns all columns*

2. What does the star (*) mean?

- The * means: give us **all the columns** in the cities table.
- It is a **shortcut**, so we do not have to write every column name.

3. Getting only some columns

We can also choose **specific columns**. For example, if we only want the name and country columns:

```
SELECT name, country FROM cities;
```

If we run that, we get only those two columns.

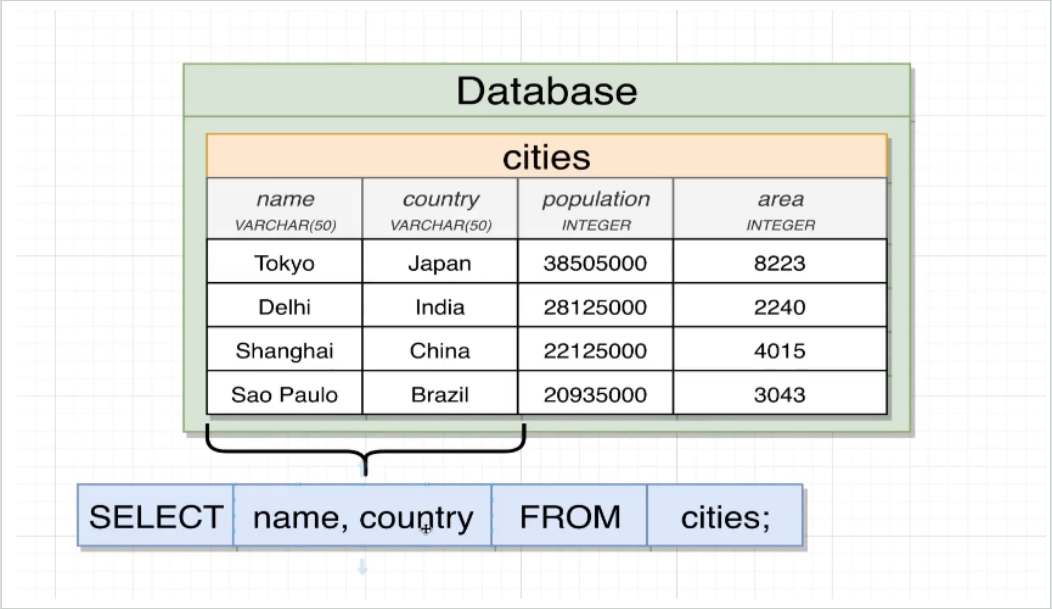


Figure 2 — SELECT name, country FROM cities; returns only two columns

More ways to list columns

Query	What we get
SELECT name, population FROM cities;	The name and population columns
SELECT name, population, area FROM cities;	Three columns. There is no limit to how many we list
SELECT area, name, population FROM cities;	We can **change the order** of the columns
SELECT name, name, name FROM cities;	We can list **one column many times**. Here we get the name column three times in a row

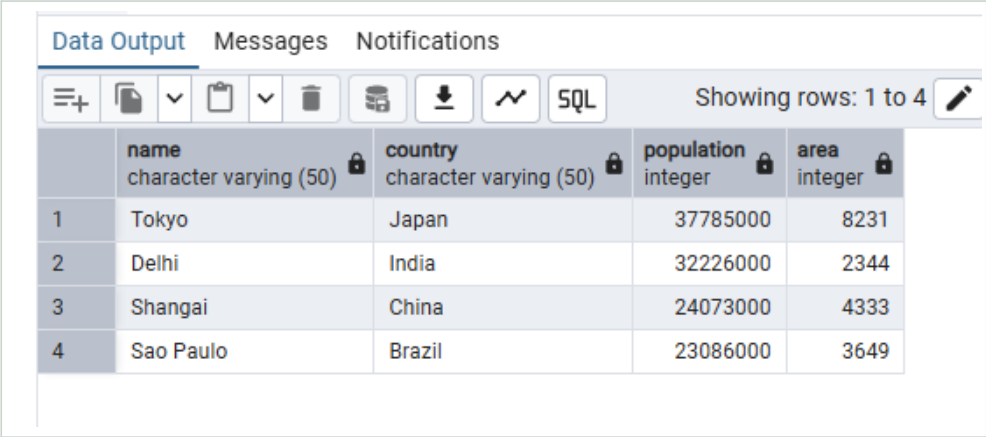
Why repeat a column? It may look pointless now, but it is sometimes useful to print the values of one column many times. We will see an example soon.

4. Running the code (my practice)

Query 1: get all the data from the cities table.

```sql
-- Show all data from the "cities" table
SELECT * FROM cities;
```

*Figure 3 — SELECT * with a comment*



Data Output | Messages | Notifications

Showing rows: 1 to 4

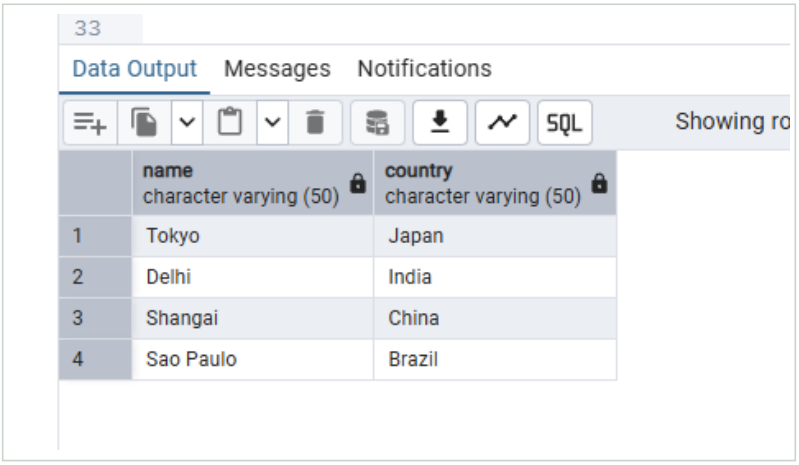
	name character varying (50)	country character varying (50)	population integer	area integer
1	Tokyo	Japan	37785000	8231
2	Delhi	India	32226000	2344
3	Shangai	China	24073000	4333
4	Sao Paulo	Brazil	23086000	3649

Figure 4 — Result: all 4 columns and 4 rows

Query 2: get only the name and country of all cities.

```sql
-- Show the name and country of all cities
SELECT name, country FROM cities;
```

Figure 5 — SELECT name, country with a comment



33

Data Output | Messages | Notifications

Showing ro

	name character varying (50)	country character varying (50)
1	Tokyo	Japan
2	Delhi	India
3	Shangai	China
4	Sao Paulo	Brazil

Figure 6 — Result: only the name and country columns

5. Summary

To pull information out of a table, remember this pattern:

```
SELECT <columns or *> FROM <table name>;
```

Part	Meaning
SELECT	List the columns we want, or * to get all of them
FROM	Then the name of the table we want information from
*	Shortcut for all columns
name, country	Only these columns. We can change the order or repeat a column

Sources
Lecture transcript and lecture images (Figures 1–2)
My own practice screenshots (Figures 3–6)